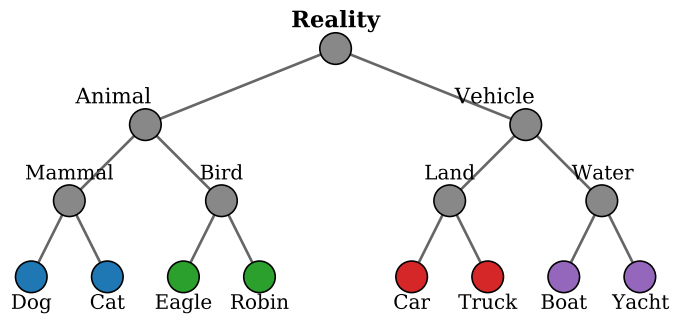
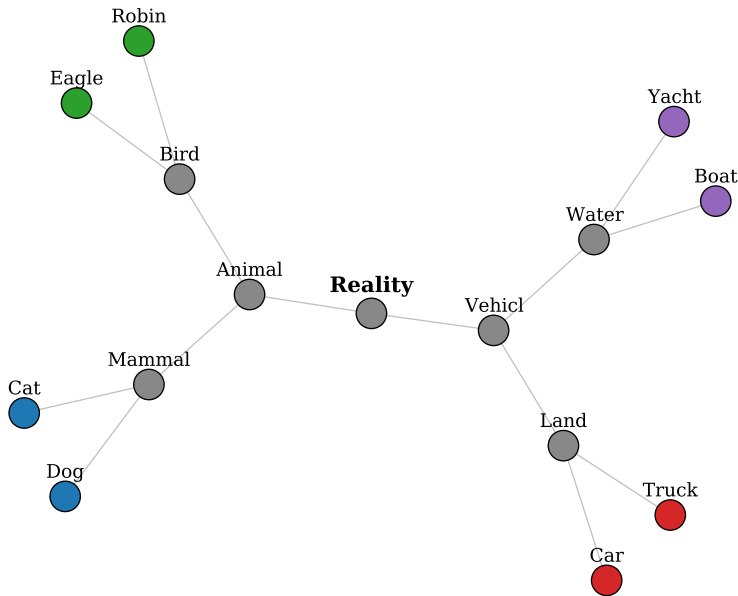


(a) A semantic hierarchy (binary tree)

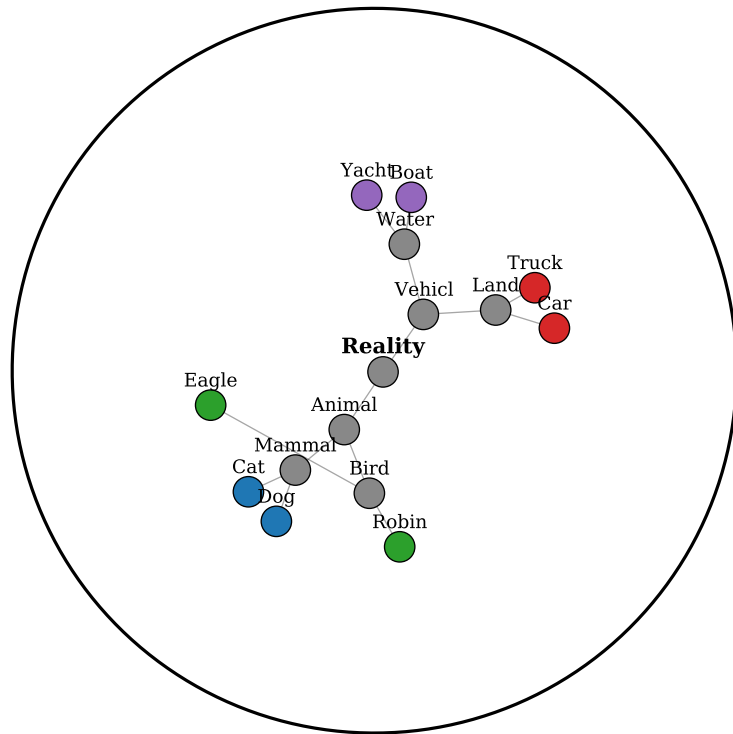


(b) Euclidean embedding ($\mathbb{R}^2$, MDS)



Branches collapse: cousins look as close as siblings

(c) Poincaré disk embedding ($\mathbb{H}^2$, Riemannian Adam)



Distances preserved: branches separate near the boundary